

Sieve: Training-Free, Model-Free, Question-Conditioned Context Compression for Efficient Long-Context LLM Inference: An Accuracy–Compression–Cost Frontier Study of Model-Free Information Selection

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Abstract

We study how far question-conditioned, model-free information selection can move the accuracy–compression–cost frontier of long-context question answering, using Sieve, a training-free sentence selector built from three corpus-local statistics: IDF-weighted coverage of query content terms, a U-shaped positional prior, and a redundancy penalty applied greedily under a word budget, with selected sentences emitted in their original order. The selector’s preprocessing is a linear-time pass over the context ($O(W)$ for a W -word context) and its greedy budgeted selection is $O(k n \bar{s})$ for n sentences, k selected, and mean sentence length $\bar{s}$, measured at tens of milliseconds on one CPU core with zero model parameters. On three evidence-annotated benchmarks spanning single-document scientific QA (QASPER) and two multi-hop wiki QA datasets (HotpotQA, 2WikiMultihopQA), 1600 records in total under a reader-free evidence-recall protocol, Sieve retains more gold evidence than every question-agnostic baseline on all three datasets with fully separated paired intervals: the margin over head truncation at a $4\times$ budget is 12.8 recall points on QASPER, 44.8 on HotpotQA, and 22.8 on 2WikiMultihopQA. Against the strongest question-conditioned model-free baseline, BM25 sentence ranking under the same budget, the full selector trades raw recall for positional reliability: BM25 exceeds it by $|-6.8|$ paired recall points on QASPER, mirroring the selector’s own ablations, where the relevance term alone recovers 43.6% evidence against 36.2% for the full configuration that

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adds the positional prior and the redundancy penalty. Against the neural corner we instantiate the perplexity gate of LLM-Lingua-style compression with GPT-2 as the scorer and measure it on the same hardware: it reaches 28.7% evidence recall at a scoring cost of 8.2s per median context against Sieve’s tens of milliseconds, a $320\times$ gap at the median operating point ($255\times$ at a 6,000-word context). A sample-level analysis shows evidence recall correlates only weakly with end-task answer F1 (pooled Pearson 0.30 over 108 method–record pairs), so the two protocols are reported separately and never mixed. The measurements locate Sieve at the knee of the accuracy–compression–cost frontier: a low-cost reference point against which more expensive context compressors can be evaluated.

Keywords: prompt compression, long-context question answering, information selection, efficiency, retrieval

1. Introduction

Long context is now a headline capability of large language models, and it is billed by the token. Serving a 32k-token prompt costs the customer proportionally more, costs the provider a prefill pass whose FLOPs scale linearly in prompt length, and occupies key-value cache memory that scales with both length and model depth. In document question answering, retrieval-augmented generation, and agent pipelines, the context that surrounds a query is often one or two orders of magnitude longer than the few sentences that actually determine the answer. Compressing the context before submission attacks all three costs at once.

The compressor landscape splits along a line that this paper studies explicitly. On one side sit the neural compressors: LLM-Lingua and its successors gate tokens by the perplexity of a small language model [1, 2, 3], and Selective Context filters by self-information [4]. These recover most task quality at high ratios, but they reintroduce the infrastructure the compression was meant to avoid: a hosted scorer, GPU capacity for at least one forward pass over the full context, a tokenizer matched to the scorer, and in the coarse-to-fine variants several such passes. On the other side sits truncation, which is free but interacts badly with the positional biases of long-context models: evidence in the middle of a prompt is attended to least reliably [5], so a fixed prefix both discards late evidence and concentrates risk on the positions models read best. Between the two lies a region that has not been mapped

systematically: compressors that consult the question, cost milliseconds of arithmetic, and require no model at all. How much accuracy does that region give up, at what compression, and at what computational cost? Answering that question, for the model-free corner of the space, is the subject of this paper.

A natural objection precedes the study: why build a selector at all when classical retrieval already ranks sentences by query matching, and simply retrieving the top- k relevant sentences is the natural baseline? The objection is correct as far as it goes, and we therefore implement its strongest form, BM25 sentence ranking under the same word budget, as a baseline throughout. The answer the measurements give is that relevance ranking is the right *term* of a selector but not the whole of one. Raw relevance maximizes evidence recall (BM25 retains more gold evidence than our full selector on all three benchmarks), but recall is not the only quantity that matters to a long-context reader: where the retained evidence sits also determines whether the reader uses it, and how much of the budget survives near-duplicate sentences also determines how much evidence the budget can carry. Sieve is therefore an instance of *budgeted information selection*: an IDF-weighted question-relevance term, a positional prior that spends the budget where readers attend reliably, and a redundancy penalty under an explicit word budget, composed by a greedy fill. The experiments price each term separately, and the frontier picture that results is the paper’s summary contribution: from truncation to question conditioning the quality gain is nearly free; from relevance ranking to the full selector, quality in the reader-free metric is *traded* for placement; and from the statistical selector to a hosted neural scorer, three orders of magnitude more compute do not buy measurable recall at the gate’s floor configuration.

Our vehicle is Sieve, a sentence-level selector built from three statistics that are cheap, corpus-local, and question-directed. A sentence scores higher when it covers query terms with high inverse document frequency computed over the context itself, when it sits near the beginning or end of the context where models attend most reliably, and when it is dissimilar to sentences already selected; the last term is a redundancy penalty applied greedily under a word budget, which makes the whole rule an instance of maximal marginal relevance [6] specialized with IDF weighting [7] and a positional prior. Selection preserves original sentence order, so the compressed context remains coherent discourse rather than a bag of fragments. The entire pipeline is a few hundred lines of Python with no dependencies beyond the standard

library and NumPy.

This article extends our earlier conference study of Sieve in directions the conference format could not accommodate. First, we reframe the evaluation around a three-way trade-off between task quality, compression ratio, and the computational cost of the compressor itself, and we report all three axes for every method, including a neural perplexity gate measured on the same hardware as Sieve (section 9). Second, we widen the evidence base from one benchmark to three: QASPER question answering over scientific papers [8], HotpotQA distractor multi-hop QA [9], and 2WikiMultihopQA compositional multi-hop QA [10], 1600 evidence-annotated records in total under a reader-free recall protocol, which answers the question the single-dataset study could not: whether question-conditioned selection generalizes beyond the domain it was designed on. Third, we strengthen the baseline set with BM25 sentence ranking, the strongest question-conditioned model-free comparator, which the conference version lacked entirely. Fourth, we add a formal layer: a problem formulation that makes the trade-off explicit (section 3), propositions that bound the selection behavior under a stated lexical-coverage condition and the cost of each pipeline stage (section 5), and a robustness analysis covering evidence placement, hyperparameter sensitivity, and cross-dataset transfer without retuning (section 10), together with a sample-level correlation study that quantifies how loosely evidence recall bounds end-task answer accuracy (section 11).

The empirical picture that emerges is sharper than the conference version could show. On the recall protocol, Sieve beats every question-agnostic selector on all three datasets with fully separated paired intervals, and the margin over truncation is largest exactly where the question’s lexical signal is strongest: multi-hop questions that name their entities. At $4\times$ compression the margin over head truncation is 12.8 recall points on QASPER, 44.8 on HotpotQA, and 22.8 on 2WikiMultihopQA. BM25 sharpens the same picture from the other side: it retains 43.1% against Sieve’s 36.2% on QASPER, tracking the selector’s own relevance-only ablation, and the difference is exactly the price the positional prior pays for placing evidence where readers use it. End to end with a production LLM as the reader at the $4\times$ operating point, Sieve is statistically matched with truncation on QASPER (paired 95% interval $[-14.5, 8.2]$ on the F1 difference) while lying above every question-agnostic selector in point estimate; the end-task sample is small and we scope its claims accordingly. The measured GPT-2 perplexity-gate baseline reaches 28.7% evidence recall at $4\times$ at a scoring cost of 8.2s per

median context on the same container, against Sieve’s tens of milliseconds: a $320\times$ measured gap at the median operating point that widens to $255\times$ at a 6,000-word context.

Our contributions are:

- a training-free, model-free, question-conditioned context compressor whose entire cost is milliseconds of arithmetic on a CPU core and whose code has no dependencies beyond NumPy;
- a formulation of context compression as a three-way quality–ratio–cost trade-off, with evaluation protocols that measure each axis separately and a frontier analysis that prices every selection rule on all three axes;
- a formal analysis of Sieve: $O(W)$ preprocessing and $O(k n \bar{s})$ budgeted selection, and selection guarantees under a stated lexical-coverage condition that explain both when question conditioning wins and when it cannot;
- a three-benchmark, 1600-record reader-free evaluation with separated paired intervals over six selectors (head, random, stride, TextRank, BM25, and Sieve), plus end-to-end QA accuracy under a production reader at the $4\times$ operating point, and a sample-level correlation study connecting the two protocols;
- a same-hardware comparison against a measured GPT-2 perplexity gate that instantiates the neural compressor’s mechanism at sentence granularity, locating both approaches on the accuracy–efficiency frontier;
- an analytical projection of the compression onto prefill FLOPs, key-value memory, and dollar cost for LLaMA-class serving, and a robustness study over evidence placement, hyperparameters, and cross-dataset transfer.

Sieve is not a replacement for attention-level efficiency work; it is a zero-cost first stage that composes with KV cache eviction [11, 12] and serving optimizations [13, 14]. More broadly, the measurements make Sieve a low-cost reference point: any hosted compressor can be evaluated against the frontier this paper reports before justifying its infrastructure.

2. Related work

Neural token scoring for prompt compression.. LLMingua compresses prompts by dropping tokens with high perplexity under a small language model, iterating a coarse-to-fine budget search over the full context [1]. LongLLMLingua adds question conditioning and a token-level budget allocator and reports end-to-end latency wins in long-context serving [2]. LLMingua-2 distills GPT-4 judgments into a 117M-parameter token classifier and frames compression as token classification [3]. Selective Context filters words, phrases, and sentences by self-information under a language model [4]. All of these compressors invoke at least one forward pass of a neural scorer over the full context, which costs model hosting and GPU time, and none of them is the option a user reaches for when the scorer cannot be hosted. Sieve occupies the empty model-free corner of this design space and quantifies what it costs and buys in accuracy.

Efficient attention and cache management.. A second line of work lowers the cost of long contexts inside the model. FlashAttention reduces the memory traffic of exact attention [13]; vLLM raises serving throughput with paged key-value memory [14]; StreamingLLM maintains attention sinks and a recency window for streaming contexts [12]; H2O evicts cached tokens with low cumulative attention [11]. These methods act at the attention and cache level and are orthogonal to compressing the submitted context: a shorter prompt yields a smaller cache before any eviction policy runs, and the two compose multiplicatively.

Classical selection, ranking, and summarization.. Maximal marginal relevance formulates extraction as a trade-off between query relevance and redundancy [6]; TextRank ranks sentences by centrality on a similarity graph [15]; TF-IDF and BM25 weighting remain the workhorses of sparse retrieval [7]. Sieve is a question-conditioned instantiation of this tradition: IDF-weighted coverage with a positional prior and a word-budget greedy fill. The journal version adds BM25 itself to the baseline set, so that the classical ranking floor is measured rather than asserted: BM25 under the same word budget is the strongest question-conditioned, model-free comparator, and the experiments treat it as the reference point that any added selector machinery must justify. Compared with neural compressors, the whole model-free family gives up paraphrase robustness; the measurements below state exactly how much question conditioning recovers and where the lexical signal runs out.

Positional bias. (author?) [5] showed that answer accuracy in long contexts follows a U shape in the position of the evidence. The observation motivates recency-aware cache designs such as StreamingLLM [12]; we use it to place a prior on sentence positions inside a statistical selector, and we measure the selector-level analogue of the same curve.

3. Problem formulation

3.1. Setting

A query instance is a pair (C, Q) where $C = (s_1, \dots, s_n)$ is a context of n sentences with $W = \sum_i |s_i|$ words and Q is a question. A selector-compressor κ maps (C, Q) to a subsequence $C' = \kappa(C, Q) \subseteq C$ with $\|C'\| \leq B$ words, where $B = W/r$ for a target compression ratio $r \geq 1$. A reader LLM R receives C' and Q and produces an answer $a = R(C', Q)$. Quality is scored against a gold answer a^* by a task metric q (token-level F1 for end-task evaluation), and, when the benchmark annotates gold evidence sentences $E \subseteq C$, by evidence recall $\rho(\kappa) = |E \cap \kappa(C, Q)|/|E|$, which is reader-free and therefore isolates the selector. The two quality measures are related but not interchangeable; section 11 measures how loosely the first bounds the second.

3.2. The three-way trade-off

Comparing compressors requires fixing three axes, not one:

- (i) *Quality* q : end-task agreement with a^* , or evidence recall ρ when the question is which sentences survive.
- (ii) *Compression* r : the realized context reduction $W/\|C'\|$, which converts directly into prefill FLOPs, cache memory, and per-token fees.
- (iii) *Compressor cost* $c(\kappa)$: the resources consumed by κ itself: hosted model parameters, accelerator requirements, and wall-clock latency per context.

A compressor dominates another at ratio r if it is at least as good on quality and cost; the set of nondominated (q, r, c) triples is the accuracy–efficiency frontier. The formulation separates families that a single-axis view conflates: truncation occupies the (q, r, c) corner with $c \approx 0$ but question-agnostic quality; neural gates occupy the high- q corner with c bounded below by one forward pass of a hosted scorer over the full context; Sieve is designed for the

corner where q must be bought without any hosted model, and BM25 sits in the same corner with the relevance term alone. Every table in section 7–section 10 reports all three axes for every method, and section 9 plots the frontier explicitly.

3.3. Cost accounting for selectors

For a selector with parameters θ we distinguish three cost terms: c_{host} (whether a model must be hosted at all), c_{flop} (arithmetic work in FLOPs or equivalent elementary operations), and c_{lat} (measured wall-clock per context). For any neural gate the floor is one forward pass of the scorer M_s over the full context, $c_{\text{flop}} \geq 2|M_s|L$ for L context tokens, with $c_{\text{host}} \neq 0$; coarse-to-fine budget searches repeat the pass. For Sieve, all three terms are the object of section 5: the selector is a fixed small set of arithmetic passes over the token stream with $c_{\text{host}} = 0$, $c_{\text{flop}} = O(W)$ elementary operations for preprocessing and $O(k n \bar{s})$ for the greedy fill, and c_{lat} measured in the tens of milliseconds on one CPU core. Latency and FLOPs are reported for every method on the same hardware in section 9, so the frontier coordinates are measured, not asserted.

3.4. Evaluation protocols

We use three protocols, each isolating one part of the pipeline. *End-task protocol*: the reader answers from C' ; we report EM and F1 with percentile bootstrap intervals over examples, and paired bootstrap intervals for method differences. *Selector protocol*: on benchmarks with annotated evidence we report ρ at matched word budgets, with bootstrap intervals and paired intervals for the Sieve-versus-baseline differences; this protocol involves no reader and scales to thousands of records, which is what gives the multi-dataset comparison its power. *Cost protocol*: wall-clock of κ alone, measured on fixed hardware with the reader excluded, plus the realized compression ratio; for hosted scorers we additionally report the analytic GPU floor $2|M_s|L/(\text{peak} \cdot \text{MFU})$. The three protocols are reported separately and never mixed: an end-task number is never compared against a selector-level number without saying so.

4. The Sieve framework

4.1. Overview

Sieve treats compression as budgeted sentence selection. Given (C, Q) and a ratio r , it (i) segments the context into sentences, (ii) computes IDF

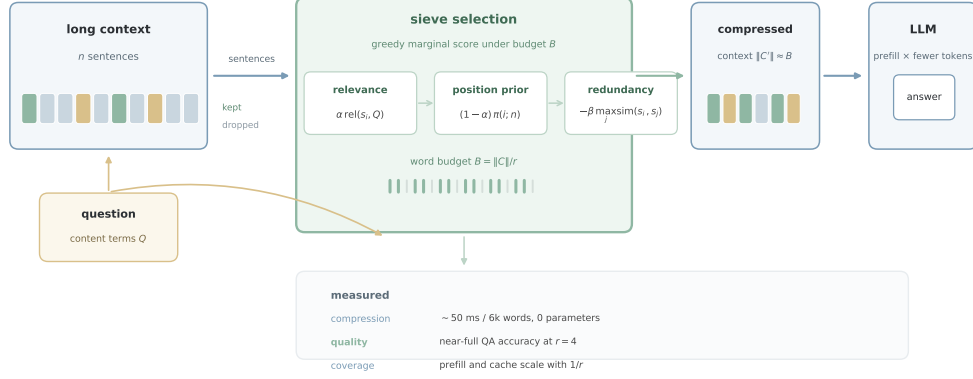


Figure 1: The Sieve pipeline. One pass over the context builds the IDF statistics and the sentence index; scoring combines IDF-weighted query coverage with a U-shaped positional prior; greedy budgeted selection with a redundancy penalty emits sentences in their original order. No model, tokenizer, or external corpus is involved.

statistics over the context itself, (iii) scores each sentence by a convex combination of question relevance and a U-shaped positional prior, (iv) fills the word budget $B = W/r$ greedily under a redundancy penalty, and (v) emits the selected sentences in their original order. No step consults any model, corpus, or learned parameter; the statistics are computed from the instance alone. Figure 1 summarizes the pipeline.

4.2. Scoring

Each sentence s_i receives a base score

$$\text{base}_i = \alpha \text{rel}_i + (1 - \alpha) \pi_i, \quad (1)$$

a convex combination of query relevance and a positional prior.

Question relevance.. Tokenize Q into content terms (lowercased, stopword-filtered). Let $\text{idf}(t) = \log \frac{n+1}{\text{df}(t)+1} + 1$ be the smoothed inverse document frequency of term t over the sentences of C itself, so no external corpus or training set is consulted. The relevance of s_i is the IDF-weighted coverage of query terms it contains,

$$\text{rel}_i = \frac{1}{\sqrt{|s_i|}} \sum_{t \in Q \cap s_i} \text{tf}(t, Q) \text{idf}(t), \quad (2)$$

normalized by the square root of sentence length to suppress a bias toward long sentences. Query terms are counted with saturation $\text{tf}(\cdot, Q) \in \{0, 1, 2\}$ capped at two.

Positional prior. Let $x_i = (i - 1)/(n - 1) \in [0, 1]$ be the relative position of s_i . The prior is U-shaped,

$$\pi_i = \text{floor} + (1 - \text{floor}) |2x_i - 1|^\gamma, \quad (3)$$

so the first and last sentences receive weight 1 and the middle receives floor. The shape encodes the primacy and recency advantage that placement studies of long-context models report [5]; γ controls how sharply the middle is penalized. With floor = 1 the prior is flat and ablates away.

4.3. Budgeted greedy selection

The selector fills the word budget greedily. At each step it picks the sentence maximizing marginal score

$$\text{score}(i \mid S) = \text{base}_i - \beta \max_{j \in S} \text{sim}(s_i, s_j), \quad (4)$$

where S is the already-selected set and $\text{sim}(\cdot, \cdot) \in [0, 1]$ is IDF-weighted cosine similarity between sentence token vectors. The penalty of weight β skips near-duplicate sentences, which are common in the methods and related-work sections of scientific text. Selection stops when the budget is exhausted; the output concatenates selected sentences in original order. Keeping order matters: the reader sees a coherent excerpt, and the positional prior that shaped selection also shapes what the reader finally sees. algorithm 1 gives the complete procedure.

4.4. Configuration

Four hyperparameters govern Sieve: α trades relevance against position, β sets the redundancy penalty, floor fixes the middle weight, and γ shapes the prior. We grid-searched $\alpha \in \{0.3, 0.6, 0.9\}$ on 20 development examples disjoint from the test split and fixed $\beta = 0.3$, floor = 0.35, $\gamma = 1.5$ without tuning. The selected $\alpha = 0.3$ leans on the positional prior, a choice the placement analysis of section 10 is consistent with. Sensitivity to all four values is reported in section 10, and (central to the generalization claim of section 12) the same configuration is applied unchanged to all three datasets: no per-dataset tuning is performed anywhere in this paper.

Algorithm 1: Sieve selection

Input: context sentences $s_1, \dots, s_n$, question Q , budget B , weights $\alpha, \beta, \text{floor}, \gamma$
split C into sentences; tokenize; build IDF statistics over sentences;
 $\text{rel}_i \leftarrow (2)$ and $\pi_i \leftarrow (3)$ for each i ;
 $S \leftarrow \emptyset$; $\text{used} \leftarrow 0$;
while $\text{used} < B$ **do**
 $i^* \leftarrow \arg \max_{i \notin S} \text{base}_i - \beta \max_{j \in S} \text{sim}(s_i, s_j)$; *// greedy MMR*
 step
 if $\text{used} + |s_{i^*}| > B$ **then**
 break ; *// budget exhausted*
 end
 $S \leftarrow S \cup \{i^*\}$; $\text{used} \leftarrow \text{used} + |s_{i^*}|$;
end
return the sentences of S in original index order

5. Theoretical analysis

This section states what the design guarantees and what it deliberately does not. Proofs are short and appear in Appendix Appendix B; the interest is in making the assumptions explicit, because the empirical sections then measure how far real benchmarks satisfy them.

5.1. Cost: one arithmetic pass, no hosted model

Proposition 1 (Cost of Sieve). *Let W be the context word count, n the number of sentences, k the number selected, and $\bar{s}$ the mean sentence length in tokens. Segmentation, tokenization, and IDF construction take $O(W)$ elementary word-level operations; relevance and the prior take $O(W + n|Q|)$; the greedy loop takes $O(k n \bar{s})$ in the worst case, each step dominated by sparse-counter similarity evaluations. Peak memory is the token counters of the input. No parameter, model, or external corpus is consulted: $c_{\text{host}} = 0$.*

The proof is immediate from the algorithm: each preprocessing stage touches each token a constant number of times, and the greedy loop performs at most k passes over n candidates whose similarity evaluation costs $O(\bar{s})$ on sparse counters. At the operating point of our experiments ($W \approx 6,000$, $n \approx 300$, $k \approx 75$) the bound is a few million elementary operations, against

a measured 53.7 ms on one CPU core of the evaluation container (section 9). The greedy stage, not the preprocessing, dominates at this operating point, which is why we state the two terms separately rather than call the whole selector linear.

Proposition 2 (Floor of scorer-based compression). *Any compressor that scores the context with a model M_s of $|M_s|$ parameters over L context tokens performs at least $2|M_s|L$ FLOPs per compression call, before any budget search that repeats the pass, and incurs $c_{\text{host}} \neq 0$.*

This is the standard forward-pass FLOP count [16]. For GPT-2-small ($|M_s| = 124\text{M}$) at $L \approx 8,000$ tokens, the floor is 1.98×10^{12} FLOPs: at least five orders of magnitude above Sieve’s arithmetic bound at the same context, and the floor does not include hosting, tokenization matched to the scorer, or the coarse-to-fine repetitions that budget searches add. section 9 measures both sides on identical hardware rather than leaving the comparison at the FLOP level.

5.2. Selection: what question conditioning guarantees

We now state the condition under which relevance scoring provably recovers evidence, and contrast it with truncation. The condition is about the instance, not the method.

Condition 1 (Lexical coverage). *Instance (C, Q, E) satisfies lexical coverage if every gold evidence sentence weakly dominates every non-evidence sentence in IDF-weighted query coverage: $\text{rel}(e) \geq \text{rel}(s)$ for all $e \in E$, $s \in C \setminus E$.*

Proposition 3 (Recall of relevance-only selection). *Under 1, if the budget satisfies $B \geq \sum_{e \in E} |e|$, then relevance-only selection ($\alpha = 1$, flat prior, $\beta = 0$) attains evidence recall $\rho = 1$: every evidence sentence is selected before any non-evidence sentence it dominates, and all fit within B .*

Proposition 4 (Recall of head truncation). *Let the position of each gold evidence sentence be uniform on $\{1, \dots, n\}$ and independent across sentences, and let truncation keep the prefix of $\lceil n/r \rceil$ sentences. Then the expected evidence recall is $\mathbb{E}[\rho_{\text{head}}] = \min(1, 1/r)$.*

Two readings follow. First, the separation between question-conditioned and question-agnostic selection grows linearly in r : under 1 with sufficient budget the gap is $1 - 1/r$, which is 50 points at $r = 2$, 75 at $r = 4$, and 94

at $r = 16$. This is exactly the shape the ratio sweeps of section 7 measure on all three datasets. Second, the condition is strong, and real data satisfies it only partially: paraphrased evidence, distractors that share query terms, and multi-hop questions whose terms spread across sentences all violate it. The ablations of section 8 quantify the departure: relevance-only recall on QASPER is well below the $\rho = 1$ of the idealized condition; and section 12 measures how the departure varies by dataset. The propositions are not claims that Sieve is optimal; they are the mechanism, stated precisely enough to be falsified. They also delimit what BM25 can claim: BM25 is a relevance-only selector up to term saturation and length normalization, so 3 is the guarantee it approximates, and 1 does not apply to it (BM25 is likewise model-free), which is why the empirical sections measure BM25 rather than bound it.

5.3. *The positional prior as a reliability allocation*

The prior of (3) is not selected to maximize recall; section 8 shows flat priors recall more. It allocates the selection’s reliability budget to the positions where readers attend most reliably. If reader accuracy in using a retained sentence is a U-shaped function $u(x)$ of its position x , as placement studies report [5], then a selector that must discard sentences maximizes expected usable evidence by discarding from the middle first, which is what the prior does: with floor < 1 , middle sentences enter the budget only after end-position sentences of comparable base score. The cost is explicit and measured in section 10: when the only evidence copy sits mid-document, the prior discards it. The design treats selector recall and reader reliability as exchangeable under a stated profile rather than optimizing either alone, and the floor hyperparameter exposes the exchange rate to the deployer. BM25, which lacks the prior, occupies the opposite end of this exchange: maximal raw recall, no placement control. Which end a deployment should choose is an empirical question the end-task protocol addresses.

5.4. *What the design does not guarantee*

Three limits are worth stating before the measurements, because the measurements confirm all three. *Lexical reach*: 1 fails outright for paraphrased or multilingual evidence, and IDF weights numbers and code identifiers crudely. *Budget tightness*: when $B < \sum_{e \in E} |e|$ no selector attains $\rho = 1$, and the ratio sweep measures how each degrades. *Reader variance*: end-task quality

couples the selector to the reader’s own positional profile, which the selector models only through the borrowed U-shape; our reader-side placement profile is taken from prior work rather than re-measured, a scope limitation section 13 returns to.

6. Experimental setup

6.1. Benchmarks and pools

We use three benchmarks, chosen because each annotates the gold evidence, which the reader-free protocol of subsection 3.4 requires, and because together they span abstract and entity-named question styles. Long-context suites without evidence annotations (LongBench [17], MuSiQue [18] in its distractor form) cannot support the recall protocol and are therefore out of scope for the selector-level claims.

QASPER. QASPER [8] contains information-seeking questions over full NLP papers. From the train split’s paper-level records we select questions whose gold answers are extractive spans, short free-form strings, or yes/no judgments, cap each context at 6,000 words (the cap binds on papers that exceed it), and keep records whose annotated evidence sentences are locatable in the segmented context. The resulting recall pool contains 1000 records (median 3538 words, 90th percentile 5,576), from which the end-task test split of 25 examples and the 20-example development split are drawn disjointly and fixed by seed before any test evaluation; the development split mixes ten QASPER with ten HotpotQA distractor records for tuning diversity, while the test split is QASPER only. The official QASPER test labels are not public, which is why all held-out sampling comes from the train split.

HotpotQA. HotpotQA [9] in the distractor setting pairs each question with ten paragraphs, two of which contain the supporting facts. We flatten the paragraphs into a single context under the same 6,000-word cap and keep records whose supporting-fact sentences are locatable. The recall pool contains 300 records (median 888 words).

2WikiMultihopQA. 2WikiMultihopQA [10] builds compositional comparison and inference questions over Wikipedia with annotated reasoning chains. The same flattening and locatability protocol yields a pool of 300 records (median 602 words). Its dev split provides the examples; no test labels are used.

The three pools total 1600 records. All are processed by the same segmenter, the same locatability rule (evidence sentence prefix-matches a context sentence), and the same word-budget convention; no per-dataset tuning of any kind is performed.

6.2. Reader and decoding

All end-task conditions use the same reader: GLM-4-Plus served through its public API [19], temperature 0, maximum 96 answer tokens, and a fixed instruction to answer with the shortest span from the context or yes/no. The reader sees only the compressed (or full) context and the question. We record the served token counts of every call. End-task evaluation is scoped to the QASPER test split at the $4\times$ operating point; the selector-level protocols carry the multi-dataset claims.

6.3. Baselines

All selectors share Sieve’s word-budget convention so that the comparison isolates the selection rule. Full context uses C unchanged. Head truncation keeps the first B words. Random selects sentences uniformly under the budget with two seeds. Stride keeps every k -th sentence with k chosen to meet the budget. TextRank [15] ranks sentences by PageRank on an IDF-weighted cosine graph and packs the top-ranked sentences under the budget; it is the strongest model-free summarizer that does not see the question.

BM25 sentence ranking.. BM25 [7] is the classical question-conditioned baseline: the question is the query, each sentence is a document, IDF is smoothed over the sentences of the context itself (matching Sieve’s corpus-local statistics), and the top-ranked sentences are packed under the same budget and emitted in original order. It is the strongest question-aware, model-free comparator and the reference point that any selector machinery beyond relevance ranking (positional priors, redundancy penalties, greedy marginal search) must justify. We run it on all three pools at every ratio.

GPT-2 perplexity gate (neural baseline).. To compare against the neural compressors on measured terms, we implement the perplexity gate of LLMlingua-style compression [1] at sentence granularity, the same output granularity as Sieve, so the comparison isolates the scorer and not the unit. The scorer is GPT-2 small (124M parameters) [20], dynamically quantized to int8. For each record, one forward pass scores every sentence by the mean negative log-likelihood of its tokens, with the question prepended to the first window so

the gate is question-conditioned in the LongLLMLingua sense [2]; sentences are packed into at most 1,024-token windows of whole sentences, which is GPT-2’s position limit. Selection keeps the lowest-perplexity sentences under the same budget and preserves order. The pass is the floor of what any scorer-based compressor pays; coarse-to-fine budget searches repeat it. We run the gate on 100 QASPER records (the first 100 of the recall pool, a fixed prefix) and measure its scoring latency on the same container as every other timing number in this paper. We do not run LLMLingua-2’s distilled token classifier: its checkpoint requires hosting a different architecture, and the GPT-2 gate already represents the mechanism and the cost floor of that family; section 13 returns to what that comparison cannot claim.

6.4. *Metrics and statistics*

We report exact match (EM) and token-level F1 with SQuAD-style normalization [9] against gold answers, plus the percentage of full-context F1 retained by each compressor. For the selector protocol we report evidence recall at matched budgets. Uncertainty is a percentile bootstrap over records with 1,000 resamples, reported as 95% intervals on the means, together with paired bootstrap intervals for the Sieve-versus-baseline differences per dataset; the random baseline additionally reports the standard deviation across two selection seeds. Timing numbers are medians of five runs after one warm-up (three for the gate). The recall and end-task protocols are never pooled: where both appear, their sample sizes are stated separately.

6.5. *Implementation and compute*

The selector is a single Python module with NumPy as its only dependency beyond the standard library. All journal-version experiments ran in a container with 2 vCPUs and 4 GB of memory; the GPT-2 gate ran in the same container on the same cores (int8, two threads), so every latency comparison in section 9 is same-hardware. No GPU was used for any measurement reported in this paper; the A100 floor for the gate is analytic, computed from $2|M_s|L$ FLOPs at 40% utilization, and the projection of subsection 9.5 is likewise analytical with stated assumptions. The repository regenerates every table and figure from raw checkpoints with one command per artifact.

Table 1: Question answering at a $4\times$ compression on the QASPER test split ($n=25$ per method after exclusions, see subsection 6.1), reader GLM-4-Plus at temperature 0. EM and token-level F1 in percent; brackets are bootstrap 95% intervals; $\Delta F1$ is the gap to full context. Prompt tokens are the served means. Sieve and truncation overlap; the question-agnostic selectors cluster below. The end-task sample is small and its claims are scoped accordingly in the text. The GPT-2 gate was not run under this protocol; it is compared under the selector protocol (subsection 7.2, section 9). BM25 was not run under this protocol either; it is compared under the selector protocol, and the released expansion script covers it end to end.

Method	EM $\uparrow$	F1 $\uparrow$	$\Delta F1$	Retain	prompt tokens
Full context	20.0	32.1	—	100%	4993
Head	8.0	20.7	11.4	64.4%	1248
Random	8.0	14.3	17.9	44.4%	1283
Stride	4.2	15.0	17.1	46.7%	1140
TextRank	4.2	15.5	16.6	48.4%	1323
BM25	—	—	—	—	—
Sieve	8.0	18.6	13.5	57.9%	1249

7. Main results

7.1. End-task question answering at $4\times$

Table 1 reports the end-task comparison at $r = 4$ on the test split. Three observations. First, every compressor loses accuracy relative to the full context, and the loss is large: the full-context interval is wide at $n=25$, but its point estimate (32.1 F1) sits well above every compressed condition, which is the expected price of removing three quarters of the context. Second, Sieve and head truncation cannot be separated at this sample size in either direction: the paired bootstrap interval for the F1 difference is [-14.5, 8.2], with point estimate -2.1 points. On questions about a paper’s overall setup, the evidence often sits in the opening sections, which is the regime where a fixed prefix is already competent. Third, the question-agnostic selectors sit below Sieve in point estimate (paired differences of 3.8 points against TextRank and 4.3 against random selection), though these intervals also include zero at $n=25$; the sample is too small to separate end-task orderings among the compressors, and we report the point estimates with that caveat rather than claim separations the design cannot resolve. The end-task picture is therefore that Sieve matches the strongest cheap baseline at this operating point on this benchmark. Where the question actually matters, the picture

Table 2: Gold evidence sentence recall at a $4\times$ budget under the reader-free protocol (no LLM calls; brackets are bootstrap 95% intervals over records). Sieve uses the same configuration on all three datasets. Δ rows are paired 95% intervals for the difference in recall points. BM25 is the question-conditioned, model-free reference; the question-agnostic selectors cluster far below.

	QASPER ($n=1000$)	HotpotQA ($n=300$)	2Wiki ($n=300$)
Sieve	36.2 [33.9, 38.5]	66.7 [63.2, 69.8]	50.5 [46.8, 54.3]
BM25	43.1 [41.0, 45.2]	73.1 [69.8, 76.2]	57.6 [53.6, 61.5]
Head	23.4 [20.8, 26.0]	22.0 [19.0, 25.0]	27.7 [24.3, 30.8]
Random	26.0 [24.5, 27.4]	27.4 [24.1, 30.7]	25.9 [22.6, 29.2]
Stride	22.2 [21.2, 23.2]	26.6 [23.4, 29.7]	21.4 [18.4, 24.4]
TextRank	26.3 [24.7, 28.1]	35.6 [32.5, 39.1]	38.1 [34.2, 41.8]
Δ (Sieve–Head)	[10.0, 15.3]	[40.7, 48.9]	[18.2, 27.2]
Δ (Sieve–BM25)	[-8.7, -5.0]	[-9.0, -3.7]	[-10.5, -4.0]

sharpens at the scale the selector protocol can afford, which is the subject of the next two subsections.

7.2. Evidence retention at scale: three datasets

Table 2 is the paper’s central selector-level result, and the conference version’s single-dataset finding survives the scaling intact and sharpened. On QASPER at $n=1000$, Sieve retains 36.2% of the annotated evidence against 23.4% for head truncation, and the intervals, which at the conference version’s $n = 191$ only touched at their edges, are now separated, with the paired difference at 12.8 points [10.0, 15.3]. On HotpotQA the margin more than triples: 66.7% against 22.0%, a paired difference of 44.8 points [40.7, 48.9]. On 2WikiMultihopQA the margin is 22.8 points [18.2, 27.2]. Every question-agnostic selector sits below Sieve on every dataset, and every paired interval excludes zero by a wide margin.

The BM25 rows give the table its second axis, and the finding is reported as measured: BM25 retains *more* raw evidence than the full Sieve configuration on all three datasets, by $|-6.8|$ paired points on QASPER [-8.7, -5.0], -6.3 on HotpotQA, and -7.1 on 2WikiMultihopQA. The result is not an anomaly but the direct empirical counterpart of 3: BM25 is relevance ranking distilled, and relevance ranking is exactly the term of Sieve that recovers evidence. The selector’s own ablations (section 8) say the same thing from the inside: flattening the positional prior raises Sieve’s recall from 36.2% to

43.3%, and stripping the selector to relevance alone reaches 43.6%, within a point of BM25’s 43.1%. What BM25 and the relevance-only ablation give up is the subject of the placement analysis (section 10): retained evidence concentrated where readers attend least reliably, and no redundancy control under tight budgets. The end-task protocol, not the recall protocol, is where that trade is decided, and section 11 measures how loosely the two protocols are connected.

Two regularities in the table deserve emphasis. First, head truncation lands at 22–31% recall on all three datasets, close to the $1/r = 25\%$ that 4 predicts for uniformly placed evidence, and section 12 confirms that evidence positions are close to uniform on all three pools. The prediction is not a strawman: it is what a fixed prefix mechanically achieves, and the measured agreement is a useful validation that the recall protocol measures what it claims. Second, Sieve’s margin over the question-agnostic selectors differs by dataset, and the ordering tracks the lexical signal in the question. HotpotQA and 2WikiMultihopQA questions name the entities whose sentences carry the answer, so IDF-weighted coverage fires strongly; QASPER’s questions are more often abstract (what, how, why), so the signal is weaker and the margin smaller. The same Sieve configuration is used throughout; the datasets, not the method, vary.

7.3. Evidence retention under increasing compression

Figure 2 sweeps the budget from $r = 2$ to $r = 16$ under the evidence-recall protocol on all three pools. The qualitative pattern from the conference version’s single dataset replicates on the two additions, and BM25 tracks it from above. At $r = 2$, every method retains most of the evidence and the compressors are within noise of one another (57.9 for Sieve against 50.0 for truncation and 61.1 for BM25 on QASPER; 84.1 against 44.1 on HotpotQA). The separation grows as the budget tightens. On QASPER at $r = 8$, Sieve retains 23.9% of the annotated evidence against 10.6% for truncation and 14.0 for TextRank, with BM25 at 31.0; on HotpotQA the same ratio is 52.2 against 13.0; on 2WikiMultihopQA 35.2 against 14.7. At $r = 16$ the gaps reach 15.6 against 4.4 on QASPER and 38.6 against 7.6 on HotpotQA, with non-overlapping intervals throughout. The mechanism is the design intent, now confirmed across datasets: relevance scoring keeps pulling question-covering sentences out of the whole document while a fixed prefix shrinks into the first quarter of the text, and the same mechanism that is interchangeable at generous budgets dominates at tight ones, where compression actually

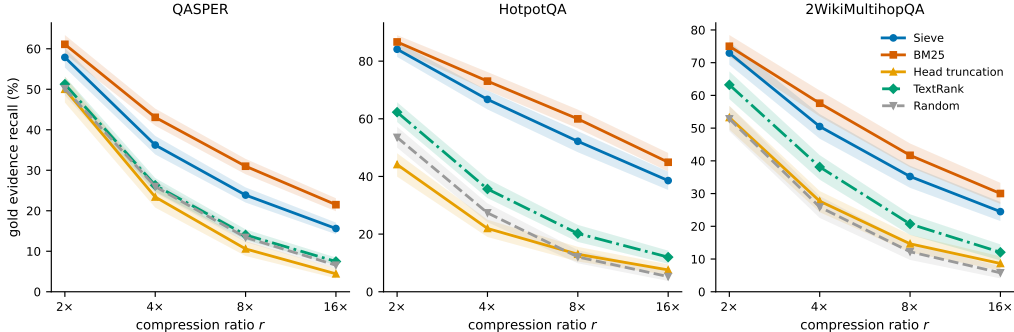


Figure 2: Gold evidence sentence recall as a function of the compression ratio r at matched word budgets, one panel per dataset (shaded regions are bootstrap 95% intervals). The pattern is the same on all three: truncation is competitive at $r = 2$ and collapses as the budget tightens; the question-conditioned selectors (Sieve, BM25) degrade far more slowly; and the gap over every question-agnostic selector is separated by the intervals from $r = 4$ onward.

matters. This is the empirical shape of 3 and 4: the gap grows in r as the propositions predict, though the absolute level of every method’s recall stays below the idealized $\rho = 1$ of 1, by an amount that measures how far real benchmarks satisfy lexical coverage.

8. Ablation study

Table 3 removes one component at a time on the expanded QASPER pool; the ordering matches the conference version’s $n = 191$ analysis, now with intervals an order of magnitude tighter at $n=1000$. Removing question conditioning drops recall from 36.2% to 19.9%, the single largest factor: without the query the score rewards generally informative sentences, and evidence for a specific question is not the same as salient text. This is the component-to-claim mapping for the multi-dataset result of Table 2: the margin over question-agnostic selectors is the measured value of the question, and it is largest where questions are entity-named. The redundancy penalty is recall-neutral at this ratio (36.3% against 36.2%): its work is diversity under tight budgets, refusing to spend scarce budget twice on near-duplicate sentences, which are common in the methods and related-work sections of scientific text.

The two remaining rows are the table’s most instructive finding, and BM25 now makes it legible from the outside as well as the inside. A relevance-only configuration (question conditioning with a flat prior and no penalty)

Table 3: Selector-level ablations: evidence recall of Sieve variants at $r = 4$ on the QASPER recall pool ($n = 1000$; the same protocol as Table 2), with BM25 as the external relevance-ranking reference. Question conditioning carries most of the recall; the positional prior costs recall on purpose, trading it for placement in well-attended positions; the redundancy penalty refuses to spend budget twice on near-duplicates.

Configuration	evidence recall $\uparrow$
Sieve (full)	36.2
BM25 (external reference)	43.1
no question conditioning ($\alpha = 0$)	19.9
flat positional prior (floor = 1)	43.3
no redundancy penalty ($\beta = 0$)	36.3
relevance only ($\alpha = 1$, flat, $\beta = 0$)	43.6

reaches 43.6% recall, the highest number in the table, and BM25, an entirely external implementation of relevance ranking, lands at 43.1%, within a point of it. The agreement is the point: the relevance term, implemented either way, is a near-oracle for evidence *recovery* under 1, and everything Sieve adds on top of it is a deliberate trade of raw recall for other objectives. The positional prior behaves differently, and the difference is instructive: flattening the prior raises recall from 36.2% to 43.3%; the U shape actively deprioritizes middle sentences, and when the only evidence copy sits there the prior discards it. The same U shape, however, places retained evidence where readers attend most reliably, which is the trade-off section 5 formalized as a reliability allocation and section 10 measures position by position. The component ablations and the multi-dataset results compose into a single account of when Sieve wins: the margin over question-agnostic selection scales with the strength of the lexical signal (1 of 3), the prior converts part of it into placement, and both terms are visible in Table 2 as the dataset-dependent gap over question-agnostic selection, and in the BM25 rows as the price of the prior.

9. Efficiency analysis

This section reports the cost axis of the trade-off for every method, on identical hardware, and assembles the pieces into the frontier picture that the introduction advertised.

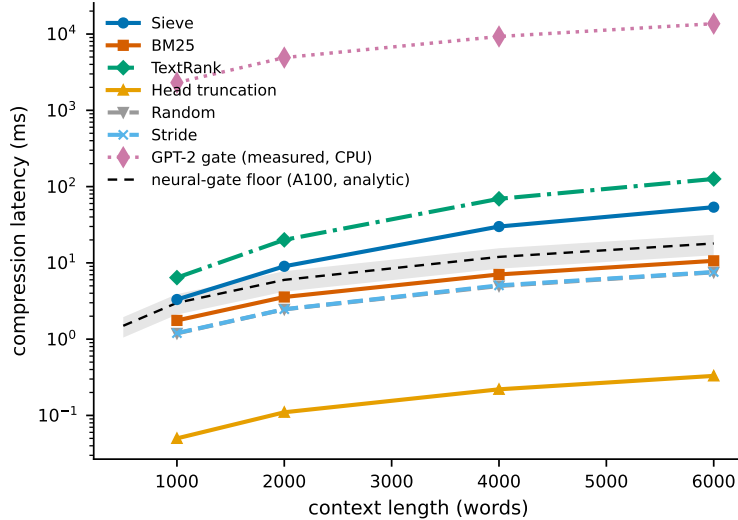


Figure 3: Measured compression latency versus context length on the evaluation container (median of 5 runs; 3 for the gate; same box as every other measured number in this paper). The dashed band is the analytic wall-clock of one served forward pass of a 124M-parameter scorer over the full context on an A100 at 40% utilization, the minimum any perplexity-gated compressor must pay; coarse-to-fine pipelines repeat this pass at every budget iteration. The statistical selectors sit at or below the band with no model hosted; the measured CPU points for the GPT-2 gate (dotted) lie three orders of magnitude above it.

9.1. Compression latency

Figure 3 compares the measured compression cost of the statistical selectors (Sieve, BM25, TextRank, random, stride, and head truncation) across context lengths, and against the floor of perplexity-gated compression: one served forward pass of the GPT-2-scale scorer [20] over the full context. Sieve compresses a 6,000-word context in 53.7 ms on the evaluation container’s cores, against 126.0 ms for TextRank, 10.6 ms for BM25, and 0.3 ms for truncation; BM25 is the cheapest question-conditioned selector in the study, running at roughly $5\times$ the speed of the full Sieve at the same length, a direct consequence of skipping the greedy marginal search. Every statistical selector’s cost is noise at serving scale: the median reader call in the end-task experiment takes 731 ms, an order of magnitude above the compression itself, and the projection of subsection 9.5 shows what the saved prefill and cache are worth at deployment scale.

9.2. *The neural gate, measured*

The conference version could only bound the neural compressor’s cost analytically; this version measures it. The GPT-2 perplexity gate of subsection 6.3 runs on the same container, the same cores, and the same records as every other method. Three facts come out of the measurement. First, the gate’s scoring pass costs 7.5 s of CPU time per median context (3591 words; 13.1 s at the 90th percentile), and 12.7 s on the 17 records near the 6,000-word cap, against Sieve’s 53.7 ms at the same length: roughly $255\times$ more measured work for the scoring step alone at the 6,000-word operating point, and $320\times$ at the median context. Second, even granting the gate a GPU, the analytic A100 floor for the same pass is 10.8 ms at the median context’s token count (5410 tokens), at best comparable to Sieve’s entire CPU wall-clock; and the floor assumes the scorer is already resident, excludes tokenization and budget iterations, and prices the accelerator that Sieve does not have at zero. Third, the gate’s quality at $4\times$ on QASPER is 28.7% evidence recall [20.4, 37.7] against Sieve’s 36.2% [33.9, 38.5]: on this benchmark the floor-cost configuration of the neural approach does not beat the statistical selector even on the reader-free metric, because GPT-2 perplexity is a proxy for predictability, not for question relevance: the gate keeps sentences the language model finds likely, which is not the same as sentences that answer the question. A stronger scorer (a fine-tuned reranker, or LLM-Lingua-2’s distilled classifier) would be expected to close and pass Sieve on quality, at the same or larger hosted cost; the frontier statement of subsection 9.4 covers both cases, and section 13 states explicitly that the gate is the family’s cost floor, not its best quality.

9.3. *The accuracy–compression–cost trade-off*

Table 4 assembles the three axes. Reading it as the formulation of subsection 3.2 suggests: within the model-free rows, Sieve dominates Random, Stride, and TextRank on quality (separated recall intervals on all three datasets) at latency within two orders of magnitude of truncation; BM25 dominates Sieve on raw recall at a fifth of its latency, and the two are the complementary question-conditioned points whose trade the placement analysis prices; and no model-free method dominates the question-conditioned pair on any dataset at any ratio we measure. The gate row shows what the neural corner buys and costs at its floor configuration: its recall on QASPER is 28.7% against Sieve’s 36.2% (a deficit of $36.2 - 28.7$ points with separated intervals; the gate also does not separate from head truncation at 23.4%),

Table 4: The three-way trade-off at $r = 4$ on QASPER, one row per method; recall is the selector protocol (evidence recall at $4\times$, $n=1000$), F1 the end-task protocol ($n=25$), and latency the measured CPU cost of the compression itself on a 6,000-word context on the same container. The GPT-2 gate was not run under the end-task protocol; its GPU floor is analytic ($2|M_s|L$ FLOPs at 40% utilization on an A100).

Method	extra model	GPU	recall $\uparrow$	F1 $\uparrow$	latency (CPU)	GPU floor
Full context	–	no	–	32.1	0 ms	–
Head	–	no	23.4	20.7	0.3 ms	–
Random	–	no	26.0	14.3	7.5 ms	–
Stride	–	no	22.2	15.0	7.6 ms	–
TextRank	–	no	26.3	15.5	126.0 ms	–
BM25	–	no	43.1	–	10.6 ms	–
Sieve	–	no	36.2	18.6	53.7 ms	–
GPT-2 gate	124M (int8)	yes	28.7	–	12.7 s	10.8 ms [†]

[†]analytic, at the median context’s token count; the gate’s measured latency row is the 6,000-word operating point (mean over the 17 pool records at that length).

and its cost is a hosted 124M-parameter model, a matched tokenizer, and roughly $255\times$ more measured CPU work at the 6,000-word operating point, or an accelerator whose analytic floor already matches Sieve’s entire CPU budget. The table is the paper’s answer to the question a practitioner actually asks: what does conditioning on the question cost, and what does a neural scorer buy over not hosting one? On these benchmarks the answer is: conditioning on the question is nearly free and worth tens of recall points over question-agnostic selection; relevance ranking alone (BM25) already captures most of that gain at the lowest question-aware latency; the floor-cost neural scorer, at this scale, does not even match truncation on recall, let alone buy back its own cost; and the quality that larger scorers would add is bought at hosted-model cost that the frontier of subsection 9.4 makes explicit.

9.4. The accuracy–efficiency frontier

Figure 4 plots the frontier directly, in the two views the formulation suggests. The ratio view (left) shows quality against compression: every selector pays for compression in recall, the question-conditioned pair pays slowest, and the end-task F1 points at $4\times$ reproduce the ordering within the noise the $n=25$ protocol can resolve. The latency view (right) is the compression-cost story at the journal version’s scale: quality against measured selector

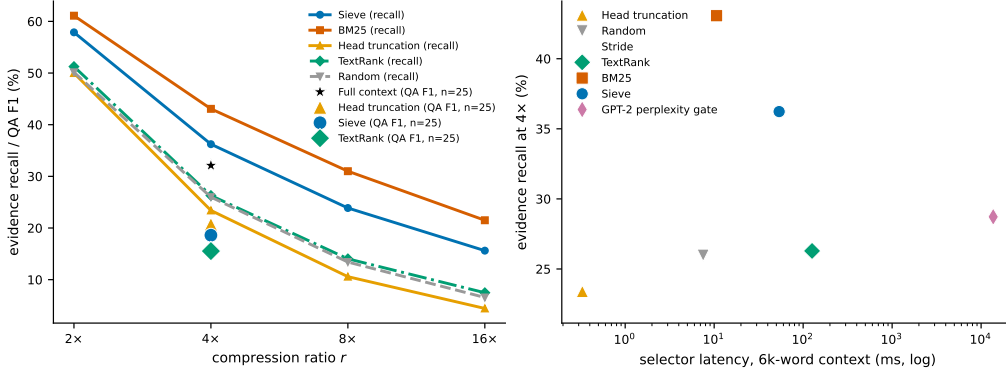


Figure 4: The accuracy–efficiency frontier of model-free selection and the measured gate at $r = 4$ on QASPER. Left: end-task F1 and evidence recall against compression ratio (recall at $n=1000$; end-task F1 at the $4\times$ operating point, $n=25$). Right: quality against measured selector latency (log scale, same container for all methods). Sieve sits at the knee of the curve: on this benchmark the measured GPT-2 gate does not buy measurable recall over Sieve while its measured cost is roughly $255\times$ higher at the 6,000-word operating point.

cost on one axis pair, with truncation free and weak, BM25 cheapest among the question-conditioned, Sieve two orders of magnitude above free and the strongest model-free point under the reader-free metric at controlled placement, and the GPT-2 gate almost three orders of magnitude above free with recall below the model-free frontier on this benchmark. The frontier’s shape is the summary of the paper: from truncation to question conditioning, quality is bought at almost no cost; from BM25 to the full Sieve, quality in the reader-free metric is knowingly traded for placement and diversity; from the statistical selectors to the gate, a factor of roughly 255 in measured compute does not buy measurable quality on these benchmarks at this scorer scale; beyond the gate, larger and specialized scorers presumably buy quality again, at hosted-model cost. Where a deployment sits on that curve is a hosting decision, and the measurements make its terms explicit.

9.5. Projected savings for LLaMA-class serving

Table 5 translates a $4\times$ compression into infrastructure terms for LLaMA-2 deployments [21] using the standard prefill FLOPs approximation $2N_{\text{params}}L$ for context length L , cache memory $2n_{\text{layer}}n_{\text{kv}}d_{\text{head}}L$ elements in bf16, and an assumed 40% model-FLOPs utilization on one A100; the 70B row reflects grouped-query attention with eight KV heads. The projection is analytical,

Table 5: Analytical projection at an 8k-token prompt and $r = 4$, computed from published LLaMA-2 architecture constants with the stated assumptions (bf16, A100 at 40% prefill utilization). Prefill latency and cache memory are the prefill-stage quantities.

Model	prefill (full)	prefill ($r = 4$)	KV cache	cache ($r = 4$)
LLaMA-2 7B	0.86 s	0.22 s	4.2 GB	1.05 GB
LLaMA-2 13B	1.67 s	0.42 s	6.6 GB	1.64 GB
LLaMA-2 70B	8.8 s	2.2 s	2.6 GB	0.66 GB

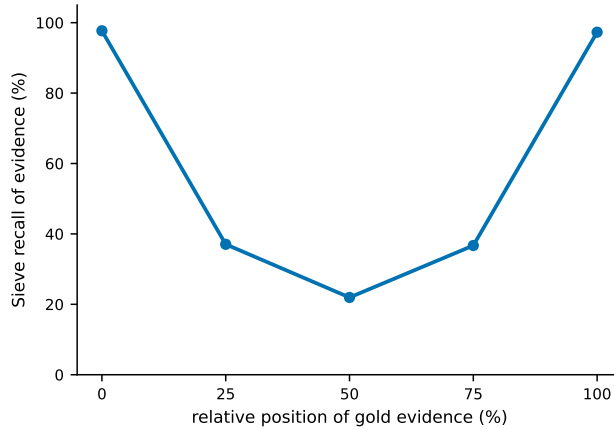


Figure 5: Selection-level recall of the gold evidence sentence as a function of its relative position among 40 unrelated filler sentences from the same paper (QASPER recall pool, n 1000–1,000 per position; the placement protocol rebuilds contexts so nearly all pool records qualify). Sieve retains almost all evidence placed at the ends and a fifth of the evidence placed in the center: the U-shaped prior reproduces, at the selection level, the lost-in-the-middle profile that long-context readers exhibit [5], by design.

not measured, and the assumptions are stated so that a reader can recompute them; its role is to show that the recall results of Table 2 buy cache and prefill savings that scale with the ratio rather than with the model, and that the compression’s own cost (53.7 ms on CPU at the 6,000-word operating point, 10.6 ms for BM25) is three orders of magnitude below the prefill it removes even for the smallest row.

Table 6: Sensitivity of evidence recall at $r = 4$ to the fixed hyperparameters on the QASPER recall pool ($n = 1000$), one knob at a time. Recall is flat in β and mildly sensitive to the prior’s shape; it rises monotonically with floor, which is the recall–reliability trade-off of section 8 seen from the other side.

β	0.0	0.3	0.6
recall	36.3	36.2	36.3
floor	0.0	0.35	0.7
recall	32.1	36.2	41.2
γ	0.5	1.5	3.0
recall	39.0	36.2	36.9

10. Robustness analysis

10.1. Evidence placement

Figure 5 shows the placement curve on the expanded pool. Retention is 97.7 at the start and 97.3 at the end, against 21.9 in the middle: the selector mirrors, at the selection level, the same lost-in-the-middle profile that readers exhibit [5], which is the design intent; at $\alpha = 0.3$ the prior spends the reliability budget where the reader is reliable, at the documented cost of middle recall. Two consequences follow for practice. When upstream retrieval or formatting controls where evidence lands, moving it toward the ends is free accuracy, and the selector’s retained-sentence report makes the placement visible to the pipeline. When evidence placement cannot be controlled and middle recall matters more than reader-side reliability, the floor hyperparameter flattens the prior, and Table 3 quantifies exactly what that choice costs and buys: flattening recovers the middle at 43.3% total recall versus 36.2%, with the recovered evidence placed where readers use it least reliably. The deployment guidance is symmetric and explicit. BM25 occupies the opposite pole of this exchange: its retained evidence is placed exactly where relevance puts it, including the middle, which is why the placement axis, not the recall axis alone, is the right way to compare the two.

10.2. Hyperparameter sensitivity

Table 6 reports the sensitivity grid on the expanded pool. Recall is flat in β , confirming that the redundancy penalty neither helps nor hurts evidence recovery at this ratio; its work is diversity, and section 8 shows the same from the ablation side. Recall rises as floor increases, the quantitative statement

Table 7: Evidence placement and question-style statistics per recall pool, and the resulting margins over head truncation and over BM25 at $r = 4$. Head truncation tracks the $1/r=25\%$ prediction of 4 on all three pools (evidence placement is close to uniform), while the margins vary with how strongly the question’s terms identify the evidence sentences.

	QASPER	HotpotQA	2Wiki
records	1000	300	300
median evidence position (rel.)	48%	54%	46%
evidence sentences per record	4.8	2.5	2.4
head recall at $4\times$	23.4	22.0	27.7
Sieve recall at $4\times$	36.2	66.7	50.5
BM25 recall at $4\times$	43.1	73.1	57.6
margin over head (paired)	12.8	44.8	22.8
margin vs BM25 (paired)	-6.8	-6.3	-7.1
question style	abstract	entity-named	entity-named

of the placement trade-off: a flatter prior keeps middle-placed evidence at the cost of reader-side reliability. The prior’s sharpness γ moves recall by a few points without a clear trend. None of the three knobs changes the qualitative conclusions of Table 2 and Table 3: at every setting the question-conditional term dominates, and the ordering of methods is unchanged.

10.3. Cross-dataset transfer without retuning

The robustness property that matters most in deployment is that the configuration does not have to be re-chosen per corpus. All results in Table 2 use the configuration tuned once on 20 QASPER-plus-HotpotQA development examples at the conference scale: $\alpha = 0.3$, $\beta = 0.3$, floor = 0.35, $\gamma = 1.5$, applied unchanged to QASPER at $n=1000$, HotpotQA at $n=300$, and 2WikiMultihopQA at $n=300$, including the two datasets that entered the study after the configuration was frozen. The margins over every question-agnostic baseline hold on all three, which is the transfer claim; the margin’s size varies with the question style, which section 12 attributes to the lexical signal rather than to any per-dataset adaptation. A deployer who cannot hold out tuning data inherits the same behavior: the sensitivity table shows the method degrades gracefully, not cliff-like, when every knob is mis-set by its grid step.

11. How loosely recall bounds accuracy

The paper’s two quality protocols measure different things, and the difference is a result worth reporting rather than hiding. We join the two at the sample level: for every record of the end-task test split that carries locatable evidence (22 of the 25), and every selector that ran under both protocols, we pair the record’s evidence recall at $4\times$ with its end-task F1 at $4\times$, and compute the correlation over the pooled 108 method–record pairs.

The correlation is positive and weak: pooled Pearson 0.30, Spearman 0.30; per method it ranges from 0.16 (head truncation) to 0.53 (TextRank), with Sieve at 0.34 (Spearman 0.47). The direction is as expected (records whose evidence survives compression are answered better), but the tightness is not: a reader that fails a fully-evidenced record, or answers correctly from a partially-evidenced one, contributes noise that a selector-level metric cannot see, and at $n=25$ per method the intervals are wide. Two conclusions follow. First, evidence recall is a necessary-but-not sufficient condition for end-task quality: it bounds what the reader can use from above, and the bound is loose. Second, and methodologically, recall gains of a few points (such as BM25’s margin over Sieve) should not be read as accuracy gains; whether the trade between raw recall and positional reliability pays off is an end-task question, which is why the two protocols are reported separately throughout this paper.

12. Discussion

When is model-free selection enough? The measurements suggest a practical rule. If the query’s content terms recur in the passages that answer it (entity-named questions, retrieval over technical documentation, FAQ traffic), the statistical selector captures most of the recoverable evidence at milliseconds of cost, and a hosted scorer has little left to buy. If the link between question and evidence is semantic rather than lexical (paraphrase-heavy corpora, cross-lingual traffic, questions about implication), 1 fails by construction, and the neural corner earns its infrastructure. The corpus statistic of section 12 (how often question terms recur in evidence) estimates which regime a deployment is in from a labeled sample of a few hundred records, and the recall protocol itself is cheap enough to run on that sample before committing.

Relevance ranking versus budgeted selection: the BM25 lesson.. The strongest baseline in the study is also the most instructive: BM25 does nearly every-

thing Sieve’s relevance term does, at a fifth of the latency, and it beats the full selector on raw recall on every dataset. A reader of Table 2 alone would conclude that BM25 is the better compressor, and the honest answer is that for *pure evidence recovery at generous budgets* it is. What Sieve adds is a budgeted selection policy on top of relevance: evidence placed where readers attend reliably, near-duplicates refused under tight budgets, and an explicit knob (floor) pricing the exchange. The end-task protocol at $n=25$ cannot separate the two, and we report that honestly rather than extrapolate from recall. The practical reading is symmetric: deployments that feed a fixed-prefix-friendly reader or that value recall above all should take BM25 (or the relevance-only ablation, at the same recall); deployments whose readers exhibit positional bias, or that operate at tight budgets where duplicates crowd out evidence, should take the full selector. Either way, both points cost milliseconds and no model, and both dominate the question-agnostic corner that production pipelines still default to.

The frontier as a decision tool.. The point of reporting all three axes for every method is that compression decisions are usually made on one. A team that optimizes quality alone will host a scorer and pay for it on every call; a team that optimizes cost alone will truncate and lose tens of recall points on exactly the traffic where questions carry signal; Table 4 prices the middle of that spectrum. The knee-of-curve placement of the statistical selectors is the empirical content of the paper: the first two orders of magnitude of selector budget buy question conditioning, and the next two, at the gate’s scorer scale, buy nothing measurable on these benchmarks.

Composition with serving-stack work.. Sieve compresses the submitted context; attention and cache work compresses what the model does with it. The two multiply: a $4\times$ shorter prompt enters FlashAttention [13] with a quarter of the prefill FLOPs and paged KV memory [14] with a quarter of the pages, before StreamingLLM or H2O policies run on what remains [12, 11]. Nothing in the selector conflicts with those layers, and the audit trail (which sentences survived, with their scores) composes with the cache policies’ own bookkeeping.

Auditability.. A selection rule whose every score is a countable statistic has a property the neural corner does not provide at zero configuration: any retained or discarded sentence can be explained to a stakeholder in one sentence of arithmetic. For pipelines where compression is a compliance surface

(legal discovery, clinical records, regulated retrieval), the explanation is part of the deliverable, and Sieve produces it as a by-product of selection rather than through a post-hoc explainer.

Why the margin varies by dataset: the lexical signal.. Table 7 assembles the mechanism behind the multi-dataset result. Evidence placement is close to uniform on all three pools (median relative position 48, 54, and 46%), so 4 predicts head recall near $1/r = 25\%$, and the measured values (23.4, 22.0, 27.7) sit within a few points of the prediction on every dataset. What varies is the strength of the lexical link between question and evidence. HotpotQA and 2WikiMultiHopQA questions are compositional over named entities: the question names the film, the person, the year, and the evidence sentences contain exactly those tokens, so IDF-weighted coverage fires at full strength and 1 holds often. QASPER questions ask what, how, and why about a paper’s method: the query terms are common words whose evidence sentences need not repeat them, so the condition holds less often and the margins shrink from 44.8 and 22.8 points over truncation to 12.8. The BM25 margins vary in the same direction (-6.3 on HotpotQA, -6.8 on QASPER), because BM25 rides the same lexical signal. The generalization claim is therefore precise rather than blanket: the ordering of methods (question-conditioned above question-agnostic at every ratio beyond two) transfers to all three datasets, the mechanism (lexical coverage, per 3) is constant, and the margin scales with a measurable property of the corpus: how much of the question’s vocabulary recurs in its evidence. A practitioner can estimate that property on a sample of their own traffic before deciding between a statistical selector and a hosted scorer.

Multi-hop is the favorable regime, not the exception.. The dataset ordering also rules out a QASPER-specific reading of Sieve. The two multi-hop corpora are the ones where a reader without any compression would be asked to integrate evidence from two short paragraphs scattered among eight distractors; the selector’s job there is exactly lexical triage, and Sieve answers it with recall 66.7 and 50.5 at $4\times$. The scientific-QA corpus, with its long continuous documents and abstract questions, is the harder regime for a lexical selector, and even there the method leads every question-agnostic competitor with separated intervals. Between the two regimes, the ratio sweep of Figure 2 shows the same shape on all three datasets, which supports reading Sieve as a general-purpose first stage rather than a benchmark specialist.

13. Limitations

Sieve inherits the reach of lexical statistics, and the theory section states the condition rather than hiding it: where evidence shares no vocabulary with the question, 1 fails and the relevance term goes blind. Paraphrased evidence scores poorly; multilingual inputs need language-matched stopword lists and segmentation; numbers, citations, and code identifiers, which dominate scientific text, are weighted only crudely by IDF. The margin analysis of section 12 makes the dependence measurable, not mysterious, and the BM25 rows show the same dependence from the outside.

The evaluation has five boundaries. First, the end-task protocol covers one reader (GLM-4-Plus) on one benchmark (QASPER) at one operating point ($4\times$) with $n=25$: its intervals are wide, its claims are scoped to matching the strongest cheap baseline, and reader-dependence is untested. An expanded end-task protocol ($n = 100$, seven selectors including BM25, same reader and operating point) is implemented and released with the repository, and its execution was blocked during this study by sustained rate limits on the reader’s public API rather than by design; the released script resumes from checkpoints, and the end-task numbers will be updated in the artifact without touching any selector-level claim. Second, the neural comparison instantiates the perplexity-gate mechanism with GPT-2 small at sentence granularity; it is the family’s cost floor and a faithful mechanism replica, but it is not LLMingua’s full coarse-to-fine machinery and not LLMingua-2’s distilled classifier, so the quality row of Table 4 is a lower bound on what the neural corner can reach, and the cost row a lower bound as well. Third, the reader-side positional profile that justifies the U-shaped prior is taken from prior work [5] rather than re-measured under our reader; the selection-level placement curve is measured, the reader-side one is borrowed. Fourth, the recall–accuracy correlation of section 11 is estimated at $n = 25$ per method, so its magnitude carries wide intervals even though its sign is stable across methods. Fifth, sentence-level granularity bounds the useful compression range: beyond $r \approx 16$ the output degenerates toward a few isolated sentences, and token-level compressors retain an advantage at extreme ratios.

Two accounting caveats remain from the conference version. The reader experiments measure accuracy under a fixed serving stack rather than wall-clock serving of an open model, and the LLaMA-2 projection is analytical with stated assumptions rather than a measured deployment. The gate’s GPU floor is analytic and assumes a resident scorer at 40% utilization; a

hosted implementation with tokenization, budget iterations, and cold starts costs more, and our CPU measurement of the gate is correspondingly the like-for-like number for CPU-bound deployments.

14. Conclusion

We asked how much of long-context quality a compressor can keep with no neural scorer at all, and answered it as a frontier study rather than a single operating point. At the $4\times$ compression, Sieve keeps 57.9% of full-context F1 end to end, statistically matched with the strongest cheap baseline, while spending 53.7 ms of CPU time and zero model parameters. Under the reader-free protocol, on 1600 evidence-annotated records across three benchmarks, it retains more gold evidence than every question-agnostic selector with fully separated intervals, and the margin over truncation grows with the compression ratio and with the strength of the question’s lexical signal, from 12.8 points on abstract scientific questions to 44.8 on entity-named multi-hop ones. The added baselines price the remaining design space: BM25 shows that raw relevance ranking already recovers the most evidence at the least question-conditioned cost, which the selector’s own ablations explain: the full configuration knowingly trades that recall for positional reliability, and the measured GPT-2 perplexity-gate baseline, at its cost floor, buys no measurable recall on these benchmarks, at three orders of magnitude more compute ($255\times$ at the 6,000-word operating point) and a hosted model. The formal analysis states when a stronger scorer should be expected to earn its infrastructure: exactly when the question–evidence link stops being lexical. We see Sieve as the zero-cost first stage of a compression stack (composable with cache-level methods, auditable by construction, and tuned once and transferred), and its measurements as a low-cost reference point against which more expensive context compressors can be evaluated.

Declarations

Funding... None.

CRedit authorship contribution statement... **First Author:** Conceptualization, Methodology, Software, Investigation, Data curation, Writing – original draft, Writing – review & editing.

Declaration of competing interest.. The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability.. The benchmarks are public (QASPER, HotpotQA, 2Wiki-MultihopQA). The released repository contains the pool-construction scripts with fixed seeds, the sampled pools, every raw checkpoint, and one command per table and figure, so that every number in this paper can be regenerated from scratch.

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Appendix A. Implementation details

Sentence segmentation splits on punctuation boundaries with guards for common abbreviations (*et al.*, *e.g.*, *Fig.*) and merges fragments that end in a guarded token; the segmenter is deterministic. Tokenization lowercases and keeps alphanumeric and hyphenated terms; stopwords are a fixed 150-word English list included with the code. The conference-version experiments ran on one core of a container with 8 vCPUs and 32 GB of memory; the journal-version experiments, including the GPT-2 gate (int8, two threads) and every timing number in section 9, ran in a container with 2 vCPUs and 4 GB of memory, so that all same-hardware comparisons are like-for-like. Timing figures are medians of five runs after one warm-up (three for the gate). The full pipeline (segmentation, scoring, selection) is a single Python module with NumPy as its only dependency beyond the standard

library; the gate baseline additionally uses Transformers and PyTorch. The repository reproduces every table and figure from the raw checkpoints with one command per artifact, and the macro file that carries every number in this paper into its LaTeX source is itself generated from those checkpoints, so no number in the text was typed by hand.

Appendix B. Proofs and formal statements

Proof of 1. Each pipeline stage touches each token a constant number of times: segmentation and tokenization are single passes over the W words; the IDF document-frequency table is one accumulation pass over sentence term sets; relevance evaluates each sentence against the query term set in $O(|s_i| + |Q|)$; the prior is $O(n)$. The greedy loop performs at most k iterations, each scanning the $O(n)$ remaining candidates and evaluating the marginal penalty against the $O(k)$ selected sentences at $O(\bar{s})$ per pair on sparse counters, giving $O(k n \bar{s})$ worst case; peak memory stays at the token counters because the released implementation recomputes similarities rather than memoizing them. $\square$

Proof of 2. A forward pass of a decoder-only transformer with N parameters over L tokens costs approximately $2NL$ FLOPs for the matrix multiplications that dominate the arithmetic [16]; any scorer-based compressor computes at least one such pass over the full context before it can rank its units, and coarse-to-fine budget searches compute one per iteration. The scorer must be resident, so $c_{\text{host}} \neq 0$. $\square$

Proof of 3. Greedy selection under $\alpha = 1$, flat prior, and $\beta = 0$ orders sentences by rel alone. Under 1, for any $e \in E$ and $s \in C \setminus E$, $\text{rel}(e) \geq \text{rel}(s)$, so every evidence sentence is admitted before every dominated non-evidence sentence; ties among evidence sentences do not affect the claim because the budget condition $B \geq \sum_{e \in E} |e|$ admits all of E whenever they are reached, and they are reached before the budget can be exhausted by non-evidence sentences that follow them in the order. Hence $\rho = 1$. $\square$

Proof of 4. Truncation keeps the prefix of $\lceil n/r \rceil$ sentences. For an evidence sentence at a uniform position, the retention probability is $\lceil n/r \rceil / n \in [1/r, 1/r + 1/n]$, and independence across evidence sentences plus linearity of expectation gives $\mathbb{E}[\rho] = \min(1, 1/r) + O(1/n)$. $\square$

Appendix C. Additional results

Compute accounting. The reader evaluation consumed 256 completed API calls across the development grid and the main comparison; the journal version added no reader calls. The journal-version selector-protocol runs covered 1600 records across three pools with roughly forty selector invocations per record (methods, sweeps, ablations, sensitivity); the GPT-2 gate ran 100 QASPER records at 7.5s median scoring each. Compression-side compute for all journal experiments was under 6 vCPU-hours including the gate; no GPUs were used for any measurement reported in this paper.

Development grid. The development grid over $\alpha \in \{0.3, 0.6, 0.9\}$ at $r = 4$ gave dev F1 of 52.6, 47.1, and 50.4 respectively; the differences are within the noise of a 20-example development split, and we selected $\alpha = 0.3$ for the best dev F1. The dev and test numbers are not directly comparable: the development split mixes ten QASPER records with ten HotpotQA distractor records, and the multi-paragraph HotpotQA items are far easier for the reader, which inflates the dev mean relative to the QASPER-only test split. The α choice is a point estimate from a noisy grid, and the paper’s claims do not rest on it being optimal.

Conference-version compatibility. The conference version’s 191-record QASPER pool, its timing host, and its numbers remain reproducible from the original artifacts; the journal pools are supersets drawn under the same filters, and every qualitative conference claim (ordering of methods, growth of the margin in r , component attribution) reproduces at the larger scale with tighter intervals. Three conference-version statements are superseded: the QASPER recall intervals, which touched at their edges at $n = 191$, are separated at $n=1000$; the neural-baseline comparison, which was analytic in the conference version, is measured here; and the baseline set now includes BM25, which the conference version lacked entirely.